DEEPAVALI

Product Requirements Document

A personalized, cinematic festival experience that transforms each visitor’s own photograph into a golden Deepavali portrait reveal.

Product	DEEPAVALI – Personalized Interactive Diwali Website
Document	Complete PRD • Version 1.0
Prepared	8 November 2026
Implementation environment	AntiGravity IDE
Deployment	Vercel

DESIGN NORTH STAR

Dark cinematic atmosphere • warm gold lighting • authentic visitor photo • human-only cutout • real-time particle bursts • polished mobile experience.

This PRD is the implementation source of truth. Where a detail has not yet been explicitly finalized, it is labelled as a proposed default for the first build and can be adjusted without changing the core experience.

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Priority key: P0 = required for launch; P1 = important; P2 = enhancement. Proposed defaults are intentionally conservative and should be confirmed during implementation.

1. Executive summary

DEEPAVALI is a personalized, interactive festival website. A visitor opens a greeting experience, optionally provides a photograph, and sees a cinematic sequence that presents the person as the emotional centre of a Deepavali celebration. The scene combines a dark festive environment, warm golden light, diyas, fireworks, a subtle duplicate ghost portrait, golden particle trails, and a final greeting.

The defining visual concept is **Ghost Portrait + Gradual Golden Reveal**. The uploaded photograph must contribute only the human subject. Original scenery—trees, walls, rooms, streets, or other background objects—must not leak into the designed scene. The website supplies its own complete festive background.

Cracker bursts are not static decorations: each burst emits a visible spray of small golden sparks that travel outward, arc or drift, glow briefly, and fade naturally. The experience must be designed mobile-first and must look and operate beautifully on phones, while also scaling to tablets and desktops.

Implementation target: Next.js, React and TypeScript, deployed on Vercel; Neon Serverless PostgreSQL provisioned through the Vercel Marketplace; Drizzle ORM; Vercel Blob for private photo/media storage; and Vercel Functions for server-side operations. Supabase is not part of this architecture.

Launch definition

- A visitor can complete the greeting experience on a modern phone without needing a desktop.
- A visitor can use an uploaded photo or continue with a non-personal fallback visual.
- The photo is isolated to a human-only transparent cutout before it is layered into the scene.
- The ghost portrait, golden reveal, cracker particle bursts, greeting and controls work reliably.
- Private media is protected from public access and is not exposed through guessable URLs.
- Admin-controlled content and access rules can be managed without editing source code.

2. Product vision and goals

Vision: Make a digital Deepavali greeting feel personal, alive and memorable—not like a static greeting card or a generic slideshow.

Primary goal: Deliver a high-quality, responsive, cinematic experience in which the visitor’s own isolated portrait becomes part of a carefully art-directed festive composition.

Secondary goals: Provide a simple entry flow, reliable photo handling, manageable administration, privacy-conscious storage, fast loading, and graceful behavior on lower-powered mobile devices.

Success measures

Measure	Launch target / definition
Experience completion	At least 90% of test users who start the flow can reach the final greeting without assistance.
Mobile usability	All core actions fit and work at 360 px viewport width and above; no horizontal page overflow.
Visual quality	Portrait cutout has clean edges; no original background remains visible in the final composition.
Animation resilience	Effects remain smooth on supported devices and automatically reduce intensity when performance or accessibility settings require it.
Privacy	Unauthenticated users cannot retrieve another visitor’s private original or processed image.
Reliability	Upload, processing and greeting flows provide clear retry and fallback states.

3. Users, roles and permissions

3.1 Visitor / student

- Opens the website from a shared link or direct URL.
- May authenticate if the deployment requires a student-only experience; the exact identity provider and eligibility rule are configurable.
- Views the welcome / launch introduction and starts the greeting.
- Can upload a photograph, review the extracted subject, and continue or retry.
- Can view the finished personalized greeting and replay the animation.
- Can download or share the final greeting only if that feature is enabled by the administrator.
- Can request deletion of their uploaded media or use a session-only mode if supported.

3.2 Administrator

- Signs in through a protected admin route; admin access is never granted by a client-side role flag alone.
- Manages greeting text, visual theme settings, event availability, and optional student access rules.
- Views operational status and high-level usage metrics, not private photographs by default.
- Can disable uploads, disable sharing, or place the site into maintenance mode.
- Can remove media or user records when authorized and can review audit events for administrative actions.

3.3 System / background services

- Validates uploads, stores originals privately, performs or invokes subject extraction, creates processed assets, and records processing status.
- Delivers time-limited access to private media and applies retention / deletion rules.
- Serves responsive UI and animation assets with reduced-motion and low-performance alternatives.

4. Experience principles and visual direction

4.1 Art direction

- **Atmosphere:** deep charcoal / near-black base, warm amber and gold light, restrained maroon accents, and high contrast for readable greeting text.
- **Portrait:** the visitor’s real person, not an AI-invented replacement. Preserve identity-relevant appearance and clothing as faithfully as the source permits.
- **Ghost layer:** a second, larger or offset duplicate of the isolated person, very low opacity and softly blended behind the primary subject.
- **Golden reveal:** the main cutout begins subdued and becomes gradually illuminated by opacity, glow, rim-light and particle effects.
- **Festive environment:** diyas, rangoli-inspired floor detail, lanterns, subtle foliage or ornament, distant fireworks and ambient bokeh belong to the website scene—not to the source photo.
- **Typography:** decorative display type for the main greeting, paired with a highly legible supporting font. Text must remain readable on small screens.

4.2 Non-goals

- Do not reconstruct a face from sparkles or rely on facial recognition to make a face shape.

- Do not preserve the original photo’s scenery as part of the final composition.
- Do not generate a new person or materially alter the visitor’s identity.
- Do not make the page a static poster only; the reveal and cracker particle behavior are core to the experience.
- Do not make animation mandatory for understanding or accessing the greeting.

5. End-to-end user journeys

5.1 First-time visitor flow

1. Visitor opens the landing page and sees the Deepavali visual identity, a short invitation, and a clear start action.
2. An optional cinematic rocket-launch / light-trail intro plays or can be skipped. The visitor is never trapped in a long intro.
3. Visitor reaches the greeting setup screen and chooses to add a photograph or continue without one.
4. If photo is selected, the visitor sees upload guidance, file requirements, consent/privacy note, and progress feedback.
5. The system validates and processes the image, extracts the person, and presents a preview of the isolated cutout.
6. Visitor confirms the cutout or retries / chooses another image. If extraction fails, a clear fallback is offered.
7. The greeting scene loads. The ghost portrait is visible faintly; golden light and particles build around the person.
8. One or more cracker bursts launch, emit radial spark particles and fade. The main portrait gradually becomes more visible.
9. The greeting text appears with a final golden flourish. Visitor can replay, download, share or exit according to enabled features.

5.2 Returning visitor

- Returning visitor can open a shared greeting link if it is designed to be shareable.
- Any persistent personalized content must be protected by an appropriate access mechanism.
- Expired or deleted media must show a graceful unavailable state rather than a broken image.

5.3 Admin flow

1. Admin signs in through the protected admin route.
2. Admin edits approved greeting copy, event state, theme configuration and feature toggles.
3. Admin previews changes before publishing.
4. System records the change in an audit log and publishes the updated configuration.

6. Functional requirements

ID / priority	Requirement	Expected behavior
FR-01 / P0	Landing page	Loads a polished introduction with a clear start action and no blocking content.
FR-02 / P0	Optional intro	Rocket / light-trail introduction can be skipped, respects reduced motion, and does not delay access.
FR-03 / P0	Photo upload	Accepts supported image formats, validates size/type, shows progress and allows retry.
FR-04 / P0	Human-only extraction	Removes the original background and returns a transparent subject cutout; no trees or scenery from the source remain.
FR-05 / P0	Preview & consent	Shows the processed result before using it in the greeting; user can accept or replace it.
FR-06 / P0	Ghost portrait	Renders a faded duplicate of the isolated subject behind the main portrait.
FR-07 / P0	Golden reveal	Transitions the main subject from faint to illuminated with a controlled, cinematic build.
FR-08 / P0	Cracker bursts	Burst event emits many small sparks that radiate outward, drift, glow and fade.
FR-09 / P0	Greeting composition	Displays editable greeting text, portrait, festive environment and final flourish.
FR-10 / P0	Mobile responsive	All screens and interactions adapt to narrow screens, touch input and portrait orientation.
FR-11 / P0	Fallback	If no photo or processing failure, displays an elegant non-personal greeting scene.
FR-12 / P1	Replay	Replays the reveal without requiring another upload.
FR-13 / P1	Download/share	Offers export or native share where enabled and supported.
FR-14 / P1	Admin console	Allows authorized admins to manage content and feature settings.
FR-15 / P1	Retention/deletion	Applies defined media expiry and deletion workflow.

7. Signature visual experience and animation specification

7.1 Scene layers (back to front)

1. Base background: deep dark gradient / scene backdrop.
2. Environmental depth: distant lanterns, soft bokeh, subtle decorative silhouettes and distant fireworks.
3. Ambient particle layer: slow, low-density golden dust with varied depth and speed.
4. Ghost portrait: same transparent human cutout, enlarged or offset slightly, low opacity (initial design range 5–12%), soft blur only if it does not damage silhouette recognition.
5. Main portrait: transparent cutout with subtle warm rim light; opacity and illumination animate over time.
6. Foreground effects: cracker burst particles, curved spark trails, occasional glints and diya flames.
7. Greeting typography and decorative diya / ornament, positioned to avoid covering the person’s face.

7.2 Suggested timeline (tunable)

Phase	Indicative timing	Visual behavior
Arrival	0–1.5 s	Dark festive scene fades in; ambient lights and distant particles are already alive.
Ghost reveal	1.0–2.5 s	Faint duplicate portrait appears behind the main cutout.
Golden build	2.0–5.0 s	Warm illumination grows around the subject; main portrait becomes recognizable.
Cracker burst	3.0–6.0 s	A bright flash / launch cue leads into outward-moving spark particles; sparks arc, drift and fade.
Greeting finale	5.0–7.0 s	Greeting text and ornament resolve into place with a soft glow.
Idle loop	After finale	Low-density particles, diya flicker and distant fireworks continue without overwhelming the portrait.

7.3 Particle behavior

- Each cracker burst is a distinct event with a launch cue, brief bright core, radial or fountain-like particle emission, and a natural fade.
- Particles vary in size, brightness, lifetime, angle, velocity and slight gravity / drag to avoid a uniform digital ring.
- Some particles may leave short glowing trails; trails must taper and fade rather than remain permanently.
- Particles should move in front of and behind the subject where depth is visually appropriate, but must not obscure the eyes or greeting text for long.
- Randomness must be seeded or constrained enough that the scene remains visually balanced and reproducible for QA.
- Limit particle count on mobile / low-power devices; preserve the visual idea with fewer, larger, carefully timed sparks.

7.4 Portrait reveal behavior

- Use the same processed cutout for both ghost and main layers to preserve exact likeness and alignment.
- Animate opacity, scale by a very small amount, warm glow / shadow and optional light sweep; avoid face warping.
- Keep face and body proportions unchanged. Do not stretch the cutout to fill a fixed box.
- Maintain a consistent focal point around the person’s face and upper body while ensuring full-body uploads can still be framed.
- Provide a crop / positioning preview when the uploaded subject’s aspect ratio differs from the scene.

7.5 Audio (optional)

If sound is added, it must be opt-in or user-initiated, start muted by default where browser policy requires, include a visible mute control, and never be necessary to understand the sequence. Avoid startling volume spikes. Respect device silent / reduced sensory preferences.

8. Photo upload, cutout and processing

8.1 Input guidance

- Recommend one clearly visible person, good lighting, adequate resolution and a pose with visible edges.
- Support JPEG, PNG and WebP where practical; reject unsupported, corrupt or zero-byte files with a clear explanation.
- Set a configurable upload size limit. Initial suggested maximum: 10 MB per image, subject to hosting and processing constraints.
- Do not require a specific camera or device. Mobile camera roll and file picker must both work.

- Warn users that very busy backgrounds, overlapping people, motion blur or low light may reduce cutout quality.

8.2 Processing pipeline

1. Client selects file and checks basic file type / size before upload.
2. Server validates file signature, size, dimensions and content type; do not trust the filename or browser MIME alone.
3. Original is uploaded to a private storage location with an opaque identifier.
4. Background-removal / segmentation service or processing module creates a transparent PNG/WebP cutout. Provider choice is an implementation decision and must be reviewed for cost, privacy, terms and reliability.
5. Post-process edges carefully: preserve hair strands and clothing details, remove obvious halos, and avoid erasing parts of the subject.
6. Store processed output privately; associate it with a user/session and processing status.
7. Return a time-limited preview URL or authenticated image route. Never expose a permanent public bucket URL for private student images.
8. After preview approval, the scene uses the processed asset. Apply retention and deletion policy to original and processed versions.

8.3 Human-only requirement (critical)

The final visual must contain only the human subject from the source photograph. Original trees, walls, furniture, buildings, landscapes, other people and scenery must be removed. The festive background is generated by the DEEPAVALI website itself. A photo that still contains a rectangular original background or visible scenery is not an acceptable successful result.

8.4 Quality checks and fallback

- Show a preview with a checkerboard or neutral transparency indicator so users can verify that the background is removed.
- Offer retry with another photo if extraction fails or leaves major background fragments.
- Where possible, allow simple user correction such as erase / restore brush or crop / reposition; treat as P1 if it materially delays launch.
- If no reliable cutout can be produced, do not silently composite the original photo. Offer a non-personal greeting or ask for another image.
- Do not use facial recognition or sparkle reconstruction as the extraction mechanism.

9. Responsive and mobile-first requirements

Mobile support is a core launch requirement, not a later enhancement. The website must look polished and remain fully usable on phones, including Android devices such as Samsung Galaxy F15, as well as iPhones and tablets.

9.1 Layout rules

- Design mobile-first at a 360 px viewport baseline; verify at 320 px where feasible, 360 px, 390 px, 412 px, 768 px, 1024 px and wide desktop.
- Use a portrait-first composition on phones: keep the person central, greeting text below or above the subject depending on available space, and avoid cramped side-by-side layouts.
- On desktop, use a wider cinematic composition with the portrait on one side and greeting typography on the other where appropriate.
- Use responsive image sizing with object-fit: contain for the person cutout; do not crop the head, hands or feet unexpectedly.
- Keep primary actions within comfortable thumb reach and use touch targets of at least 44 × 44 CSS pixels.

- Prevent horizontal scrolling, clipped controls, overlapping text and unsafe-area collisions.
- Account for mobile browser address bars, dynamic viewport height, notches and bottom safe-area insets.
- Keep the upload, preview, replay and share controls usable in portrait orientation; support landscape gracefully.

9.2 Mobile performance

- Use responsive image variants and modern formats; avoid sending desktop-sized assets to phones unnecessarily.
- Reduce particle count, blur radius, shadow complexity and simultaneous animation layers on lower-powered devices.
- Pause or reduce background animation when the tab is hidden; resume safely when visible.
- Use requestAnimationFrame for canvas effects and clean up animation loops on unmount.
- Do not make the whole page depend on heavy video or large animation libraries to render the first screen.
- Ensure the page remains usable if GPU-heavy effects are disabled.

9.3 Mobile QA checklist

- Test upload from camera roll and file picker on Android Chrome and iOS Safari.
- Test touch gestures and ensure there are no hover-only controls.
- Check that portrait is not hidden behind greeting text, browser chrome or the bottom action bar.
- Check reduced motion, low battery / low-performance mode where testable, and device rotation.
- Verify download and native share behavior on mobile, with clear fallback if unsupported.
- Check text wrapping, font loading, keyboard visibility and form controls on small screens.

10. Admin and content management

10.1 Admin capabilities

- Protected sign-in and role-checked admin routes.
- Edit greeting title, supporting copy, event dates and optional attribution.
- Enable / disable uploads, audio, sharing, replay and intro animation independently.
- Configure theme presets and safe ranges for particle density, animation duration and portrait opacity.
- Preview desktop and mobile layouts before publishing.
- View aggregate operational metrics and processing failure counts without exposing personal images unnecessarily.
- Manage retention settings and handle deletion requests.
- Review an audit trail for content changes, access-control changes and deletion actions.

10.2 Admin security

- Admin authorization must be enforced server-side on every protected operation.
- Do not rely on hidden buttons or client-side route guards as security.
- Use least-privilege roles and secure session handling.
- Record actor, action, target identifier, timestamp and outcome for sensitive changes.

11. Technical architecture

11.1 Approved stack

Layer	Technology	Responsibility
Web application	Next.js + React + TypeScript	Responsive pages, routing, components, scene rendering and client interactions.
Hosting / deployment	Vercel	Production hosting, previews, environment variables and serverless execution.
Server operations	Vercel Functions / Next.js server routes	Authentication checks, upload signing, validation, processing orchestration and data access.
Database	Neon Serverless PostgreSQL via Vercel Marketplace	Users / sessions, greeting configuration, media metadata, processing state and audit records.
ORM	Drizzle ORM	Typed schema, migrations and safe database access.
Private media	Vercel Blob	Private originals and processed cutouts, subject to access-control configuration.
Animation	CSS + Canvas / requestAnimationFrame; optional lightweight motion library	Portrait transitions, particle systems, glow, fireworks and responsive effects.
Testing	Unit, integration, end-to-end and device/browser testing	Verify core flows, security, responsive layout and performance.

11.2 Architecture flow

Step	Flow
1	Browser → Next.js application on Vercel
2	Browser → authenticated upload endpoint / upload authorization
3	Server validates request and stores original in private Vercel Blob
4	Server invokes selected background-removal / segmentation capability
5	Processed transparent cutout stored privately; metadata and status saved in Neon
6	Browser receives authorized preview access and asks visitor to confirm
7	Scene renders the approved cutout with ghost layer, golden reveal and particle effects
8	Optional export/share uses a composed output generated through a controlled route or client-side renderer.

11.3 Architecture constraints

- Do not reintroduce Supabase. Neon is the approved database and Vercel Blob is the approved private media storage direction.
- Keep secrets and provider credentials on the server; never ship them to browser bundles.
- Use Drizzle migrations as the schema change source of truth.
- Keep image processing behind a well-defined adapter so the provider can be changed without rewriting the greeting experience.
- Prefer server-side authorization and short-lived access for private media.
- Keep the visual scene decoupled from the upload provider: it should consume a processed transparent image asset and scene configuration.

12. Data model and storage

Entity	Important fields (illustrative)	Purpose
users	id, email / external_subject, role, created_at, status	Visitor identity and role; only store required identity data.
sessions	id, user_id or anonymous_token_hash, expires_at, created_at	Secure visitor session and temporary anonymous flow.
greeting_configs	id, title, subtitle, theme_json, enabled_features, published_at, updated_by	Editable greeting content and visual settings.
media_assets	id, owner_id/session_id, blob_key, asset_type, mime_type, size_bytes, width, height, status, created_at, expires_at	Private original and processed media metadata.
processing_jobs	id, media_asset_id, provider, status, error_code, started_at, completed_at	Track segmentation and processing lifecycle.
share_records	id, owner_id, token_hash, media_asset_id, expires_at, revoked_at	Optional controlled sharing; use opaque, revocable tokens.
audit_logs	id, actor_id, action, target_id, metadata_json, created_at	Admin and security-relevant activity.

Storage rules

- Store private media with opaque keys; do not use names, email addresses or student IDs in storage paths.
- Keep original uploads separate from processed transparent cutouts.
- Use short-lived signed access or an authenticated proxy for private images.
- Delete originals and derived assets according to retention policy; deletion must include database metadata and blob objects.
- Do not place uploaded personal photos in public static assets, source control, logs, analytics payloads or public demo fixtures.
- Use synthetic / consented test photos in development and QA.

13. Privacy, security and safety

- Explain why the photo is requested, how it is processed, where it is stored, who can access it and when it is deleted.
- Obtain clear consent before upload and before any optional public sharing.
- Do not make a visitor’s image publicly discoverable by default.
- Validate file type by content, enforce size / dimension limits, and reject malformed files.
- Protect upload and admin endpoints with authentication / authorization, rate limits and abuse controls.
- Use HTTPS, secure cookies, CSRF protections where applicable, safe headers and server-side input validation.
- Do not log raw images, full private URLs, tokens or sensitive personal information.
- Restrict admin media access to legitimate support / moderation needs and record access where appropriate.
- Provide a clear deletion mechanism and define how quickly media and derived versions are removed.
- Review the chosen background-removal provider’s retention, training, data location, subprocessors and privacy terms before production use.

Suggested retention default

Proposed starting policy: temporary visitor uploads expire automatically after a short configured period (for example, 24–72 hours) unless the product explicitly requires longer retention. Final retention duration must be selected by the project owner and disclosed in the privacy notice. The implementation must make the duration configurable.

14. Performance, accessibility and compatibility

14.1 Performance goals

- Prioritize a fast first meaningful render; load non-essential fireworks and heavy visual assets after the main interface is usable.
- Lazy-load below-the-fold assets and use optimized responsive images.
- Set explicit dimensions / aspect ratios to prevent layout shifts.
- Target good Core Web Vitals on representative mobile networks and devices; measure rather than assume.
- Use a graceful low-effects mode for devices that cannot sustain the full particle scene.

14.2 Accessibility

- Respect prefers-reduced-motion and provide a visible way to skip the intro / replay.
- Do not flash rapidly or use strobing effects; avoid intense repeated full-screen flashes.
- Ensure text contrast and readable font sizes, especially over fireworks.
- All interactive controls must be keyboard reachable with visible focus states and meaningful labels.
- Do not communicate upload or processing status through color alone.
- Provide accessible names for icon-only buttons and alt text for meaningful images.
- Keep the greeting understandable if animations are paused or unavailable.

14.3 Browser support

Support current stable Chrome on Android, Safari on iOS, Chrome / Edge / Firefox on desktop, and Safari on macOS where practical. Define and test a minimum browser baseline during implementation. Unsupported features must degrade gracefully.

15. Analytics and observability

- Track anonymous or pseudonymous funnel events: landing viewed, intro skipped / completed, upload started, upload succeeded / failed, cutout approved / retried, greeting completed, replayed, downloaded and shared.
- Track processing latency, failure rate, file-size distribution and storage errors without collecting image contents.
- Use privacy-conscious analytics and avoid sending names, email addresses, raw filenames or private media URLs.
- Capture server errors with request correlation IDs and safe structured logs.
- Create operational alerts for elevated upload failures, processing failures, database errors and storage errors.

16. Error states and edge cases

Scenario	Required response
Unsupported file	Explain accepted formats and let visitor choose another file.
File too large	Show size limit and suggest a smaller image; do not silently compress in a destructive way.
No person detected	Explain that the system could not isolate a person; allow retry or continue without a photo.
Multiple people detected	Ask visitor to choose / upload a single-person photo, or provide a supported subject-selection step.
Background remnants	Show preview and retry / correction option; never claim successful isolation when scenery remains obvious.
Upload interrupted	Preserve recoverable state where safe and provide retry.
Processing provider unavailable	Show retry and non-personal fallback greeting; do not block the entire site.
Expired media	Show a friendly expired greeting state and explain that the image needs to be uploaded again.
Unauthenticated private asset request	Return unauthorized / not found without revealing whether another user's asset exists.
Small screen / slow device	Use compact layout and reduced particle mode; preserve essential portrait and greeting.
Reduced motion enabled	Skip or simplify animation while preserving the final greeting composition.
Share API unavailable	Offer download or copy-link fallback if sharing is enabled.

17. Acceptance criteria

The release is ready only when all P0 criteria pass.

- Visitor can reach the greeting from the landing page and skip the intro.
- Visitor can upload a supported photo from a phone and see clear progress.
- Original image background is removed; only the human subject is used in the composition.
- Ghost and main portrait layers align exactly and preserve the subject's proportions.
- Main portrait gradually becomes visible through golden illumination.
- Cracker bursts visibly emit sparks that radiate outward, drift, glow and fade.
- Final greeting text is readable and does not cover the person's face.
- On a 360 px wide phone viewport, the page has no horizontal overflow and all controls are usable.
- Portrait composition adapts to mobile without unexpected clipping; desktop composition remains polished.
- Reduced-motion mode and low-effects mode remain functional.
- Private originals and cutouts cannot be accessed by another visitor without authorization.
- Upload and processing failures show actionable recovery paths.
- Admin-only operations are protected server-side.
- Media deletion removes the associated private blobs and metadata according to the retention policy.

18. Delivery plan and testing

Phase	Scope	Exit condition
Phase 1 – Foundation	Next.js / TypeScript setup, Vercel deployment, Neon connection, Drizzle schema, environment configuration.	App deploys; database migrations run; secrets are server-only.
Phase 2 – Core experience	Landing page, responsive shell, greeting setup, intro / skip controls, editable content.	Visitor can navigate the basic experience on mobile and desktop.
Phase 3 – Photo pipeline	Upload validation, private Blob storage, segmentation adapter, transparent cutout preview, retry / fallback.	Human-only cutout flow passes quality and privacy tests.
Phase 4 – Signature animation	Ghost portrait, golden reveal, fireworks / cracker bursts, particle trails, diyas and greeting finale.	Animation matches visual direction and has mobile / reduced-motion modes.
Phase 5 – Admin & governance	Admin authentication, content configuration, feature flags, audit log, retention and deletion.	Admin controls work securely and media lifecycle is verified.
Phase 6 – QA & launch	Cross-device tests, performance tuning, accessibility, security review, production monitoring.	All P0 acceptance criteria pass; release checklist signed off.

Testing matrix

- **Unit:** file validation, permission helpers, scene configuration, animation state transitions and data access functions.
- **Integration:** upload authorization, Blob storage, database persistence, processing provider adapter, deletion workflow.
- **End-to-end:** complete visitor flow, no-photo fallback, processing failure, replay and admin publish flow.
- **Visual QA:** portrait cutout edges, ghost opacity, golden reveal, particle burst density, typography contrast and responsive composition.
- **Device QA:** Android Chrome including Samsung Galaxy F15-class device, iOS Safari, tablet and desktop.
- **Security QA:** cross-user asset access, expired tokens, unauthorized admin actions, upload abuse and secret leakage.

19. Risks and mitigations

Risk	Impact	Mitigation
Background removal quality varies	High	Preview before composition; allow retry / correction; test hair, dark clothing and complex backgrounds.
Processing provider cost / limits	Medium–High	Use adapter architecture, rate limits, file limits, usage monitoring and a fallback scene.
Mobile GPU performance	High	Adaptive particle counts, reduced blur, pause hidden tabs, test on real mid-range phones.
Privacy exposure of student photos	High	Private storage, short-lived access, server-side authorization, retention and deletion controls.
Animation distracts from greeting	Medium	Use hierarchy, particle restraint, face-safe zones and user-controlled reduced motion.
Cross-browser differences	Medium	Test key browser/device combinations and provide CSS / canvas fallbacks.
Intro delays access	Medium	Make intro skippable, short and non-blocking; remember skip preference only if appropriate.

20. Assumptions and open decisions

The following are intentionally presented as implementation defaults or decisions to confirm—not as already-approved facts:

- **Authentication:** confirm whether all visitors must sign in, whether access is limited to students, and which identity provider / college account system is available.
- **Greeting content:** confirm exact headline, supporting message, sender / institution branding and language options.
- **Background removal provider:** select after evaluating accuracy, pricing, privacy, latency and commercial terms. The UI and data model must remain provider-independent.
- **Sharing:** confirm whether visitors can share a public link, download an image/video, or keep the greeting private.
- **Retention:** confirm how long originals and processed images should remain stored.
- **Admin ownership:** identify authorized administrators and the process for account recovery.
- **Audio:** decide whether optional festive sound effects or music are included; audio must remain user-controlled.
- **Final output format:** decide whether export is a still image, short video, shareable interactive page, or a combination.

Implementation instruction for AntiGravity

Build from this PRD in small, verifiable phases. Start with the responsive foundation and core data model; then implement the upload / cutout pipeline; then the signature scene and animation; then admin, privacy controls and QA. Do not replace the approved stack, do not use the uploaded sample photo as a production asset, and do not fake the human cutout by placing the full original photograph over the festive background.

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